

Individual Report Railway Scheduling

Paul Tennigkeit
paul.tennigkeit@uni-potsdam.de

University of Potsdam, Potsdam 14469, Germany
<https://www.uni-potsdam.de/en/university-of-potsdam>

1 Contributions

1.1 Ideation

I had the idea of encoding energy costs, which we agreed to be our topic. As these costs depend a lot on the speed of a train, I introduced the idea of encoding variable speed. I also had the idea of using heuristics instead of optimization in the acceleration encoding to reduce solving time.

1.2 Programming

We agreed to use my encoding for reaching a goal as a baseline encoding for our team. I also coded all of the acceleration and energy cost logic, including heuristics in the acceleration encoding.

1.3 Debugging

As I did the programming, I was mainly responsible for the debugging too.

1.4 Presentation

I presented the progress of our encodings in the weekly meetings. In the final presentation I talked about our example environment results and conclusions.

2 Challenges

As it is hard to foresee the impact of encoding or environment changes on the solving time, I had problems to find an environment for different encodings, which kept solving time reasonable while still being complex enough to show different behaviors. Also the maxtime has a huge impact on solving time and the environments we could generate were not applicable to temporary speeds in that regard (as their maxtimes were too low). I had to run a lot of tests to determine the lowest maxtime that is still high enough to keep the environment satisfiable.

3 Recommendations

As raw outputs of clingo runs are hard to read, I would recommend to use some sort of visualization to support that. Also, future students should definitely look into heuristics if they have an encoding without optimization. Even a simple heuristic using basic domain knowledge can reduce the time to find the first satisfying model by more than 50 percent (just make sure to use the `--heuristic=domain` flag).